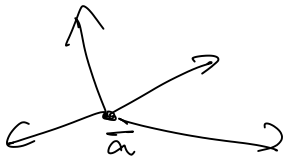
Lecture 32

Wednesday, 14 October 2026 8:24 AM

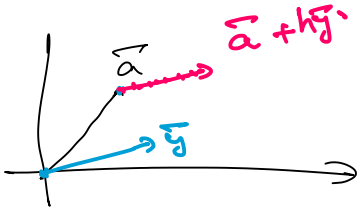
Recap: $\lim_{\vec{x} \to \vec{a}} f(\vec{x}) = L$

If $|f(\vec{x}) - L| \to 0$ as $\|\vec{x} - \vec{a}\| \to 0$

- f is cont at $\vec{a}$ if $\lim_{\vec{x} \to \vec{a}} f(\vec{x}) = f(\vec{a})$
- Derivative of a scalar field $f: \mathbb{R}^n \to \mathbb{R}$ at $\vec{a} \in \mathbb{R}^n$



Def: $$f'(\vec{a}; \vec{y}) = \lim_{h \to 0} \frac{f(\vec{a}+h\vec{y}) - f(\vec{a})}{h}$$



$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

a $a+h$

Eg1: $f(x, y) = 2x + 3y$

$$f'\left((0,0); (1,1)\right) = ?$$
$\vec{a}$ $\vec{y}$

$$f'\left((0,0); (1,1)\right) = \lim_{h \to 0} \frac{f((0,0) + h(1,1)) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{f((0,0) + (h,h)) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{f(h,h) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{(2h+3h) - (0)}{h}$$

$$= \lim_{h \to 0} \frac{5h}{h} = 5$$

Eg:- $f(x,y) = x^2 + y^2$

$f'((0,0); (0,1)) = 0$?

$$f'(\bar{a}; \bar{y}) = \lim_{h \to 0} \frac{f(\bar{a}+h\bar{y}) - f(\bar{a})}{h}$$

$f'((0,0); (1,0)) = 0$? ✓

Directional Derivative :

$f'(\bar{a}; \bar{y})$ with $\|\bar{y}\| = 1$.

$$f'\left(\underset{\bar{a}}{(x,y)}; \underset{\bar{y}}{(1,0)}\right) = \lim_{h \to 0} \frac{f(x+h, y) - f(x,y)}{h}$$

$$\frac{\partial f}{\partial x}(x,y) = f'((x,y), (1,0)) = \lim_{h \to 0} \frac{f(x+h,y) - f(x,y)}{h}$$

$$\frac{\partial f}{\partial y}(x,y) = f'((x,y); (0,1)) = \lim_{h \to 0} \frac{f(x,y+h) - f(x,y)}{h}$$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

↓ gradient of

Directional derivative : $f'(\bar{a}; \bar{y})$ with $\|\bar{y}\| = 1$

(OR)

Def. Directional derivative of f at $\bar{a}$ in

$\vec{y}$ direction is $\nabla f(\vec{a}) \cdot \frac{\vec{y}}{\|\vec{y}\|}$

Here, make sure that $\vec{y}$ has unit length.

Eg1): Find the directional derivative of $f(x,y) = x^2+y^2$ at $(0,0)$ in $(1,1)$ direction.

Method 1 | Method 2

Method 1: $f'(\vec{a}; \vec{y})$ with $\|\vec{y}\|=1$

$$f'\left((0,0); \frac{(1,1)}{\|(1,1)\|}\right) = f'\left((0,0), \frac{(1,1)}{\sqrt{2}}\right)$$
$$= f'\left((0,0), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)\right)$$
$$= 0$$

Method 2: $\nabla f(\vec{a}) \cdot \frac{\vec{y}}{\|\vec{y}\|}$

$$\nabla f(\vec{a}) = \left(\frac{\partial f}{\partial x}(\vec{a}), \frac{\partial f}{\partial y}(\vec{a})\right)$$
$$= \left(\frac{\partial f}{\partial x}(0,0), \frac{\partial f}{\partial y}(0,0)\right)$$
$$= (0,0)$$

$$D.D = \nabla f(0,0) \cdot \frac{(1,1)}{\sqrt{2}}$$
$$= (0,0) \cdot \frac{(1,1)}{\sqrt{2}} = 0$$

DD. of f at $\vec{a}$ in $\vec{y}$ direction is

$$\nabla f(\vec{a}) \cdot \frac{\vec{y}}{\|\vec{y}\|}$$

For which $\vec{y}$, D.D. is maximum?

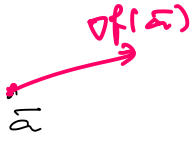
$\vec{p} \cdot \vec{q} = |\vec{p}||\vec{q}| \cos\theta$

If $\vec{y}$ is in the same direction as $\nabla f(\vec{a})$ then D.D. is maximum.



$\therefore$ The direction of Max change at $\vec{a}$ is $\nabla f(\vec{a})$.

Amount of such change is $\|\nabla f(\vec{a})\|$.



$$f(x,y) = \begin{cases} \frac{xy}{} & (x,y) \neq (0,0) \end{cases}$$

Eg:- $f(x,y) = \begin{cases} x^2+y^2 & (x,y) \neq (0,0) \\ 0 & \end{cases}$

$$\frac{\partial f}{\partial x}(0,0) = ?$$

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(h,0) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{0 - 0}{h} = 0$$

Ex:- $\frac{\partial f}{\partial y}(0,0)$, $\frac{\partial^2 f}{\partial x^2}(0,0)$, $\frac{\partial^2 f}{\partial y^2}(0,0)$